NEUROSURGERY
A-LEVEL-STYLE CORE GUIDE

30-page learning resource · Brain, spine, nerves and surgical principles

FOUNDATIONS · Anatomy, pathways, CSF and blood supply

CONDITIONS · Tumours, aneurysms, hydrocephalus, epilepsy and spinal disease

CLINICAL THINKING · Examination, imaging, risk and treatment choices

REVISION · Case studies, exam questions and answers

FORM & FIELD · SCHOLAR STUDIO BUNDLE

02 | How to use this guide

A student-friendly A-level-style introduction to neurosurgery: anatomy, physiology, common conditions, investigations, surgery, ethics and revision. This is educational material, not medical advice or a substitute for your exam-board specification. Study each condition using: definition → mechanism → symptoms/signs → investigations → treatment options → risks and follow-up. Explain cause and effect rather than memorising lists. Sudden one-sided weakness, speech difficulty, sudden severe headache, reduced consciousness, or a seizure lasting five minutes or repeated seizures without recovery needs urgent emergency assessment; call 999 in the UK.

03 | What neurosurgeons do

Neurosurgeons assess disorders of the brain, spinal cord, spine, nerves and their coverings. Examples include biopsy or resection of selected tumours, treatment of some aneurysms, CSF diversion for hydrocephalus, spinal decompression, and selected emergency operations for bleeding or pressure. Many neurological conditions do not require surgery. Care often involves neurosurgeons, neurologists, neuroradiologists, anaesthetists, specialist nurses, oncologists and rehabilitation teams. Neurology mainly treats nervous-system disorders medically; neurosurgery focuses on conditions that may benefit from an operation. Their work overlaps. Decisions balance likely benefit, risk, alternatives, uncertainty, quality of life and the person's priorities.

04 | Nervous-system overview

The central nervous system (CNS) consists of the brain and spinal cord; the peripheral nervous system (PNS) includes nerves and ganglia outside the CNS. Sensory pathways carry information towards the CNS; motor pathways carry commands towards muscles and glands. The somatic system is involved in body sensation and skeletal-muscle movement; the autonomic system regulates heart rate, digestion, pupil size and sweating. Symptom patterns can help localise a problem: weakness in an arm and leg may suggest a brain pathway problem, a sensory level may suggest spinal-cord involvement, and radiating pain may suggest a nerve root. These are clues, not diagnoses.

05 | Brain lobes and function

Frontal lobe functions include planning, judgement, behaviour and voluntary movement; the primary motor cortex lies in the precentral gyrus. The parietal lobe processes body sensation and spatial information; the primary somatosensory cortex lies in the postcentral gyrus. Temporal regions contribute to hearing, memory and language comprehension. The occipital lobe is central to visual processing. Functions arise from connected networks rather than isolated points. Language is usually strongly left-hemisphere dominant, particularly in right-handed people, but individual variation exists. Before some operations, teams map language and movement functions to reduce risk. A motor-cortex lesion may cause opposite-side weakness because many motor fibres cross.

06 | Deep brain and brainstem

The thalamus relays and processes many sensory signals. The hypothalamus contributes to temperature, thirst, hunger, circadian rhythms and hormonal control. Basal ganglia networks regulate movement; the hippocampus helps form new memories; the amygdala contributes to emotional learning. The cerebellum coordinates movement, balance, timing and motor learning; damage may cause ataxia. The brainstem consists of the midbrain, pons and medulla. It contains pathways and nuclei involved in eye movements, swallowing, arousal, breathing and cardiovascular control. Compact brainstem regions contain many vital pathways, so a small lesion may affect several functions. Surgery in deep or eloquent areas needs careful planning.

07 | Spinal cord and nerves

The spinal cord lies within the vertebral canal and is protected by vertebrae, meninges and cerebrospinal fluid. In adults it usually ends around L1–L2; nerve roots continue as the cauda equina. Dorsal roots carry sensory information into the cord; ventral roots carry motor output away. A dermatome is a skin region mainly supplied by one spinal nerve root. Peripheral nerve damage can cause weakness, numbness, tingling or neuropathic pain. Spinal-cord compression can cause weakness, altered sensation, gait difficulty and bladder or bowel dysfunction. New saddle numbness, difficulty urinating or loss of bladder/bowel control with back pain can indicate cauda equina syndrome and needs emergency assessment.

08 | Meninges, ventricles and CSF

The CNS is covered by dura mater, arachnoid mater and pia mater. The subarachnoid space contains cerebrospinal fluid (CSF) and major vessels. CSF is produced mainly by choroid plexuses, circulates through ventricles and the subarachnoid space, and returns to venous blood. It cushions the CNS and supports chemical stability and waste clearance. Simplified route: lateral ventricles → interventricular foramina → third ventricle → cerebral aqueduct → fourth ventricle → subarachnoid space. Hydrocephalus is an abnormal accumulation of CSF, often due to obstruction or impaired absorption. Treatment may include a shunt or endoscopic third ventriculostomy (ETV), depending on cause and anatomy.

09 | Blood supply and blood–brain barrier

The brain needs continuous oxygen and glucose delivery. Arterial supply comes mainly from internal carotid and vertebrobasilar systems connected through the Circle of Willis. Middle cerebral arteries supply much of the lateral hemisphere; anterior cerebral arteries supply much of the medial frontal and parietal regions; posterior cerebral arteries supply much of the occipital region. Territories vary with collateral circulation. The blood–brain barrier includes tight junctions between brain capillary endothelial cells, supported by pericytes and astrocytes. It regulates movement of substances from blood into brain tissue. This protects the CNS but makes delivery of some medicines difficult. The barrier is selective, not absolute.

10 | Neurones and action potentials

At rest, ion gradients and membrane permeability create a negative intracellular potential. The sodium–potassium pump maintains gradients by moving three Na^+ out and two K^+ in per ATP. When threshold is reached, voltage-gated Na^+ channels open and Na^+ enters, causing depolarisation. Na^+ channels then inactivate and K^+ channels open; K^+ leaves and repolarises the membrane. A refractory period limits immediate re-firing. Myelin increases conduction speed through saltatory conduction between nodes of Ranvier. Axon diameter also affects conduction speed. These mechanisms link A-level biology to clinical problems involving peripheral nerves, spinal pathways and brain function.

11 | Synapses and neurotransmitters

At a chemical synapse, an arriving action potential opens presynaptic calcium channels. Calcium entry promotes vesicle fusion and neurotransmitter release. The transmitter crosses the synaptic cleft and binds postsynaptic receptors, changing cell activity. The signal ends through reuptake, enzymatic breakdown or diffusion. Drugs can affect release, receptors, reuptake or breakdown and thereby alter signalling. Synapses allow neural networks to integrate and modify information. A-level answers should explain the sequence clearly and connect each step to the resulting response. Synaptic transmission is one reason the nervous system can combine electrical signalling along neurones with chemical communication between cells.

12 | Motor, sensory pathways and reflexes

Sensory pathways carry information towards the CNS; motor pathways carry commands to effectors. Many pathways cross sides, so brain lesions may produce opposite-side symptoms. Upper motor neurones originate in the brain or brainstem and influence lower motor neurones. Upper motor neurone damage may cause weakness, increased tone and brisk reflexes after the acute phase. Lower motor neurones connect to skeletal muscle; damage may cause weakness, reduced tone, reduced reflexes and muscle wasting over time. A reflex arc typically includes a receptor, sensory neurone, CNS synapse or interneurone, motor neurone and effector. Clinical examination combines power, tone, reflexes, sensation, coordination and gait.

13 | Neurological history and examination

A focused history establishes onset, progression, associated symptoms, trauma, medication, medical history and functional impact. Sudden onset can suggest a vascular event; gradual progression may suggest a growing lesion, but there are exceptions. Examination domains include cognition, cranial nerves, motor bulk/tone/power, reflexes, sensation, coordination and gait when safe. Cranial-nerve assessment may include vision, pupils, eye movements, facial movement and sensation, hearing, swallowing and tongue movement. Patterns help localise a lesion and select investigations. For example, language difficulty with right-sided weakness may suggest a left-hemisphere process in a person with typical language dominance. No single finding establishes a diagnosis.

14 | CT, MRI and vascular imaging

CT uses X-rays to create cross-sectional images. It is fast and useful in emergencies to detect acute haemorrhage, skull fractures, hydrocephalus and major mass effect, but uses ionising radiation. MRI uses magnetic fields and radiofrequency signals, providing excellent soft-tissue detail without X-rays. Different sequences highlight different tissue properties; contrast may help characterise tumours or inflammation. CT angiography, MR angiography and catheter angiography assess vessels. Imaging must be interpreted alongside symptoms and examination; a scan finding may be incidental, and tissue analysis may be needed to classify a tumour. NICE recommends structural MRI as the initial test for suspected glioma unless MRI is contraindicated.

15 | Brain tumours

A brain tumour is an abnormal growth in or around the brain. Primary tumours begin in the CNS; secondary tumours (metastases) spread from cancer elsewhere. Tumours may be benign or malignant, but even a non-cancerous mass can harm by pressing on vital structures. Examples include gliomas, meningiomas, pituitary-region tumours and metastases. Symptoms depend on location, size, growth and swelling: possible features include seizures, progressive headache, weakness, speech or visual changes, cognitive changes, nausea or balance problems. These symptoms have many other causes. Assessment may include neurological examination and MRI. Biopsy or resection can provide tissue for pathology and molecular testing.

16 | Brain tumour surgery

The aim of tumour surgery is to obtain diagnostic tissue and/or remove as much tumour as can safely be removed while preserving function. “Maximum safe resection” recognises that removing more is not always better if it causes permanent disability. A biopsy obtains tissue when resection is unsuitable. A craniotomy temporarily opens the skull to access a lesion. Image guidance and functional mapping help the team understand anatomy. Selected patients may have an awake craniotomy so language or movement can be tested near critical areas. Risks include bleeding, infection, seizures, CSF leak, swelling, stroke-like injury or new neurological deficit. Recovery may include rehabilitation, radiotherapy, chemotherapy or surveillance.

17 | Aneurysms and haemorrhage

A cerebral aneurysm is a weakened area of an artery that bulges outward. Many small aneurysms never rupture. Rupture can release blood into the subarachnoid space and is life-threatening. It may cause a sudden severe headache reaching maximum intensity rapidly, sometimes with vomiting, neck stiffness, collapse, seizures or reduced consciousness. Suspected subarachnoid haemorrhage is an emergency. Treatment may include microsurgical clipping or endovascular coiling. Choice depends on anatomy, rupture status, patient factors and specialist expertise. Supportive care monitors for rebleeding, hydrocephalus and delayed cerebral ischaemia. A sudden, severe, unusual headache warrants emergency assessment.

18 | Stroke and raised pressure

Stroke results from interrupted blood supply (ischaemic stroke) or bleeding (haemorrhagic stroke). Treatment differs, so rapid assessment and imaging are essential. The skull is a rigid space containing brain tissue, blood and CSF. A mass, bleeding, swelling or blocked CSF flow can raise intracranial pressure, threatening blood flow and function. Possible features include worsening headache, vomiting, reduced alertness or new neurological deficits, but presentation varies. Management treats the cause and may involve critical care, medication, CSF drainage or surgery in selected cases. Procedures such as evacuation of selected haematomas or decompressive craniectomy are major interventions with significant risks and require specialist decisions.

19 | Hydrocephalus

Hydrocephalus is an abnormal accumulation of CSF due to disrupted flow or absorption. A ventriculoperitoneal shunt drains CSF from a ventricle to the abdomen; a valve helps regulate flow. Endoscopic third ventriculostomy creates an opening in the floor of the third ventricle to bypass selected obstructions. Suitability depends on cause and anatomy. Shunts can block, become infected, drain too much or too little, or need revision. Symptoms of failure may include headache, vomiting, sleepiness, worsening balance or return of previous symptoms. Suspected malfunction needs urgent clinical assessment. Management considers cause, anatomy, pressure, age and expected benefit.

20 | Spinal disc disease and compression

Disc degeneration or herniation can irritate or compress a nerve root, producing radiating pain, altered sensation or weakness. Radiculopathy means nerve-root dysfunction; myelopathy means spinal-cord dysfunction; spinal stenosis is narrowing of the canal or nerve exit spaces. Imaging findings must match the clinical picture. Many episodes of back pain improve without surgery. Management may include activity advice, physiotherapy and clinician-selected medication. Surgery may be considered for persistent symptoms, significant neurological deficit or urgent compression. Decompression relieves pressure; stabilisation may be considered in selected cases. New bladder/bowel dysfunction, saddle numbness or rapidly progressive weakness with back pain needs emergency assessment.

21 | Spinal trauma

Spinal trauma can damage vertebrae, discs, ligaments, roots or the spinal cord. Primary injury is immediate mechanical damage; secondary injury includes later swelling, reduced blood flow and inflammation. Initial care prioritises airway, breathing and circulation while protecting the spine when indicated. Neurological examination assesses movement and sensation; imaging evaluates fractures and neural compression. Stabilisation may be needed for an unstable injury; decompression may be considered when neural structures are compressed. Rehabilitation supports mobility, independence, bladder/bowel function, skin care, respiratory health and psychological adjustment. Cervical injuries may affect arms, legs and breathing; lower injuries may primarily affect legs and pelvic-organ function.

22 | Epilepsy and surgery

A seizure is a transient episode caused by abnormal excessive or synchronous neuronal activity. Epilepsy is a tendency to recurrent unprovoked seizures; not every seizure means epilepsy. Assessment may include history, witness description, EEG and brain imaging. A normal EEG does not exclude epilepsy. Anti-seizure medicines are the main treatment for many people. If seizures remain uncontrolled despite appropriate medication, a specialist centre may assess surgery or other interventions. Options depend on seizure origin and risk to important functions. During a seizure, protect the person from injury, time it, do not restrain them and put nothing in their mouth. Seek emergency help for a seizure lasting five minutes or more, repeated seizures without recovery, serious injury, breathing difficulty or a first seizure.

23 | Parkinson's and functional neurosurgery

Parkinson's disease is associated with loss of dopamine-producing neurones in the substantia nigra and wider network changes. Symptoms can include slowness, rigidity, tremor and balance problems. Medication and rehabilitation are central. Deep brain stimulation (DBS) places electrodes in selected brain targets connected to an implanted pulse generator. Stimulation modulates neural-network activity; it does not cure Parkinson's disease and is not suitable for everyone. It may improve selected motor symptoms in carefully chosen patients. Risks include bleeding, infection, device problems and stimulation-related changes in speech, mood or balance. Programming and multidisciplinary follow-up are important.

24 | Peripheral and cranial nerves

Peripheral nerves carry motor, sensory and autonomic information. They can be injured by trauma, compression, tumours or disease. Carpal tunnel syndrome is median-nerve compression at the wrist; surgery may release the transverse carpal ligament when appropriate. Nerve repair or grafting can treat selected injuries, while decompression relieves pressure around an entrapped nerve. Nerve recovery is often slow because axons may need to regrow; function may not fully return. Nerve conduction studies and electromyography (EMG) can help assess nerve and muscle function. Tingling can originate in a nerve, nerve root, spinal cord, brain or systemic condition, so localisation comes before choosing treatment.

25 | Planning and consent

Before surgery, the team confirms the diagnosis, purpose, expected benefit, alternatives and risks of operating or not operating. Planning may include history, medicines, allergies, examination, imaging, anaesthetic assessment, positioning, navigation, monitoring and recovery. Consent is a process, not just a signature. Risks should be explained in context, including uncertainty and alternatives. A person may value independence, speech, memory or pain relief differently; those priorities should inform the discussion. Children and young people should be involved in age-appropriate ways under relevant law and guidance. Recommend an intervention because expected benefits justify its burdens and risks for that individual, not simply because it is technically possible.

26 | Anaesthesia and theatre safety

Neurosurgery commonly uses general anaesthesia, although selected procedures use awake techniques. Anaesthetists manage consciousness, pain relief, ventilation, circulation and physiological stability. The brain is sensitive to oxygen delivery, carbon dioxide, blood pressure, temperature and glucose. Monitoring may track oxygen saturation, heart rhythm, blood pressure and ventilation. Selected operations use neurophysiological monitoring to assess pathways. Navigation uses pre-operative images to orient the surgeon, although brain shift can reduce accuracy. Microscopes, endoscopes and specialised instruments improve visualisation. Safety also depends on checklists, sterile technique, teamwork, clear communication and preparation for complications.

27 | Recovery and rehabilitation

After neurosurgery, monitoring focuses on neurological status, pain, wound condition, fluid balance, seizures, infection and procedure-specific risks. Some patients need intensive or high-dependency care. Recovery goals include detecting complications early, managing pain and nausea, supporting mobilisation, nutrition and sleep, and planning wound care, medication, imaging and follow-up. Rehabilitation may involve physiotherapists, occupational therapists, speech and language therapists, neuropsychologists and specialist nurses. Recovery can be physical, cognitive and emotional and may continue for months. Fatigue, memory, mood, identity, work, education and relationships can be affected. Follow-up should address quality of life as well as recurrence or progression.

28 | Risks and complications

Risks depend on the condition, approach, patient health and urgency. Possible complications include bleeding, infection, CSF leak, seizures, swelling, stroke-like injury, new weakness or sensory loss, speech/vision/memory/balance difficulties, anaesthetic complications, blood clots, persistent symptoms or further surgery. Risk reduction includes careful selection, imaging and planning, sterile technique, physiological monitoring, communication, appropriate clot prevention and early recognition. No measure removes all risk. A complication does not automatically mean care was negligent; some occur despite appropriate treatment. For exams, connect each risk to a mechanism: injury to a motor pathway can cause weakness; disruption of a CSF barrier can cause a leak.

29 | Research and innovation

Neurosurgery evolves with improved imaging, instruments, biology and evidence. Laboratory studies explore mechanisms but do not prove benefit in humans. Observational studies identify associations but can be confounded. Randomised trials compare treatments when feasible; systematic reviews and guidelines synthesise evidence, depending on quality and relevance. Examples of innovation include improved MRI, image-guided surgery, intra-operative monitoring, endoscopic approaches, stereotactic radiosurgery and molecular tumour classification. Each tool has uses and limitations. A new technique should be compared with current standard care. Outcomes include survival, disability, symptoms, quality of life, complications, cost and access. Ask: does evidence show it works, is better than alternatives, or only technically possible?

30 | Ethics and inequality

Ethical practice includes autonomy (respect informed choices), beneficence (aim for benefit), non-maleficence (avoid preventable harm) and justice (fair access). Clinicians should explain diagnosis, uncertainty, options and likely outcomes in understandable language. Interpreters and accessible information may be needed. Family can support, but the patient's wishes remain central where they can decide. Access to specialist centres, transport, rehabilitation and social support differs between communities and may influence outcomes. Ethical reasoning weighs evidence, uncertainty, personal goals, burdens and fair use of limited resources. Strong essays apply principles to a concrete decision and explain tensions between competing values.

31 | Case study: suspected tumour

Fictional scenario: a 46-year-old develops progressive headaches and a first seizure. MRI shows a lesion near an area involved in language. Symptoms prompt assessment but do not prove a tumour. MRI helps define location, swelling and relation to surrounding structures. The team considers different tumour types and non-tumour causes. A specialist multidisciplinary team discusses biopsy, resection, monitoring or another treatment. If surgery is considered, the team weighs diagnostic value and tumour control against language impairment risk. Tissue diagnosis may change treatment. Language mapping may affect the approach. “Maximum safe resection” means removing as much as is beneficial while protecting critical function.

32 | Case study: sudden symptoms

Fictional scenario: a person suddenly develops facial droop, arm weakness and difficulty speaking. This pattern is concerning for stroke and needs emergency assessment. Clinicians establish the time last known well, assess vital functions and glucose, and perform urgent imaging to distinguish haemorrhage from ischaemia. Treatment depends on diagnosis, timing, imaging, contraindications and specialist assessment. Most acute ischaemic strokes are not treated with open neurosurgery; selected patients may receive endovascular thrombectomy. Some haemorrhages or swelling may require neurosurgical procedures. FAST: Face drooping, Arm weakness, Speech difficulty, Time to call 999 in the UK.

33 | Case study: spinal compression

Fictional scenario: a patient has neck pain, increasing difficulty with buttons, unsteady walking and new hand weakness. This raises concern for cervical myelopathy. History explores progression, trauma, pain, sensation, dexterity and bladder/bowel symptoms. Examination assesses power, tone, reflexes, sensation, coordination and gait. MRI may show cord compression. The team weighs severity, progression, imaging and risks of observation or intervention. When appropriate, decompression relieves pressure; stabilisation may be required in selected cases. Persistent compression can cause lasting injury. New bladder/bowel dysfunction, saddle numbness or rapidly worsening weakness needs emergency assessment.

34 | A-level links and glossary

Biology links: ion gradients underpin action potentials; synapses use neurotransmitters and receptors; myelin affects conduction; hypothalamic and autonomic pathways support homeostasis; respiration and circulation supply neuronal energy; altered cell signalling can contribute to tumours. Glossary: axon—process conducting action potentials; CSF—fluid around brain and spinal cord; craniotomy—skull opening usually replaced; craniectomy—bone may be left off temporarily; lesion—abnormal tissue or damage; mass effect—pressure/displacement from a lesion; resection—surgical removal; shunt—device diverting fluid; ataxia—impaired coordination; radiculopathy—nerve-root dysfunction; myelopathy—spinal-cord dysfunction. Learn terms by explaining their clinical significance.

35 | Exam questions and answers

Questions: 1) Why can left motor-cortex damage cause right-sided weakness? 2) Explain myelin's role. 3) Compare CT and MRI. 4) Why may MRI not fully classify a tumour? 5) Name two CSF-diversion procedures. 6) Why might maximum safe resection be preferable to removing every visible part? Indicative answers: many corticospinal fibres cross in the lower medulla; myelin enables saltatory conduction; CT is fast for acute blood and fractures but uses X-rays, MRI gives soft-tissue detail; tissue/molecular analysis may be needed; shunt and ETV; removing tissue near critical networks may cause irreversible harm. Longer answers should discuss monitoring versus surgery, MDT care, ethics, rehabilitation and uncertainty.

36 | Sources and disclaimer

Further reading: NHS Brain tumours — https://www.nhs.uk/conditions/brain-tumours/ ; NHS Brain aneurysm — https://www.nhs.uk/conditions/brain-aneurysm/ ; NHS Hydrocephalus — https://www.nhs.uk/conditions/hydrocephalus/ ; NICE

Brain tumours (primary) and brain metastases, NG99 — https://www.nice.org.uk/guidance/ng99 ; NICE recommendations — https://www.nice.org.uk/guidance/ng99/chapter/recommendations ; Cancer Research UK brain tumour surgery — https://www.cancerresearchuk.org/about-cancer/brain-tumours/treatment/surgery ; AQA A-level Biology — https://www.aqa.org.uk/subjects/biology/a-level/biology-7402/specification/subject-content/organisms-respond-to-changes-in-their-internal-and-external-environments-a-level-only . Educational summary only, not medical advice, a clinical guideline or a complete A-level course. Neurosurgical decisions must be made by qualified professionals; exam-board curricula differ, so check your specification.